Medical Data History Project Report

1. Project Overview

This project analyzes a medical history database using SQL to understand patient information and hospital admission records. It covers patient demographics, allergies, diagnoses, attending doctors, and province details. SQL queries are used to explore the database, examine table structures, measure dataset size, and check data quality. This analysis provides a clear understanding of the available medical data before carrying out further analysis.

2. Dataset Summary

Dataset Measure	Details
Database Name	project_medical_data_history
Total Tables	4
Total Records	Patients: 4,530 Admissions: 5,067 Doctors: 27 Province Names: 13
Total Columns	Patients: 10 | Admissions: 5 | Doctors: 4 | Province Names: 2
Database Tool	MySQL Workbench

Table and Attribute Summary

Table Name	Column Names
patients	patient_id, first_name, last_name, gender, birth_date, city, province_id, allergies, height, weight
admissions	patient_id, admission_date, discharge_date, diagnosis, attending_doctor_id
doctors	doctor_id, first_name, last_name, specialty
province_names	province_id, province_name

3. Exploratory Data Analysis Using SQL

The exploratory analysis began by checking the available databases and tables, inspecting the structure of all four tables, and confirming the dataset size. Data-quality checks were then performed to identify NULL values and repeated identifier values. The following sections document these initial checks and their outputs.

3.1 Database and Table Exploration

The available databases were displayed using SHOW DATABASES. The project database, project_medical_data_history, was selected, and its four tables were displayed using SHOW TABLES.

Database
information_schema
project_medical_data_history

Figure 1. Available databases displayed using SHOW DATABASES.

Tables_in_project_medical_data_history
admissions
doctors
patients
province_names

Figure 2. Tables available in the selected project database.

3.2 Table Structure

The DESCRIBE statement was used to inspect the fields and MySQL data types in each table. The patients table contains 10 columns, admissions contains 5, doctors contains 4, and province_names contains 2.

Field	Type	Null	Key	Default	Extra
patient_id	int(11)	NO	PRI	NULL	
first_name	varchar(150)	YES		NULL	
last_name	varchar(150)	YES		NULL	
gender	char(1)	YES		NULL	
birth_date	date	YES		NULL	
city	varchar(150)	YES		NULL	
province_id	char(2)	YES		NULL	
allergies	varchar(150)	YES		NULL	
height	decimal(3,0)	YES		NULL	
weight	decimal(4,0)	YES		NULL	

Figure 3. Structure of the patients table displayed using DESCRIBE patients.

Field	Type	Null	Key	Default	Extra
patient_id	int(11)	NO	PRI	NULL	
admission_date	date	NO	PRI	NULL	
discharge_date	date	YES		NULL	
diagnosis	varchar(50)	YES		NULL	
attending_doctor_id	int(11)	YES		NULL	

Figure 4. Structure of the admissions table displayed using DESCRIBE admissions.

	Field	Type	Null	Key	Default	Extra
▶	doctor_id	int(11)	NO	PRI	NULL	
	first_name	varchar(130)	YES		NULL	
	last_name	varchar(130)	YES		NULL	
	specialty	varchar(125)	YES		NULL	

Figure 5. Structure of the doctors table displayed using DESCRIBE doctors.

	Field	Type	Null	Key	Default	Extra
▶	province_id	char(2)	NO	PRI	NULL	
	province_name	varchar(30)	NO		NULL	

Figure 6. Structure of the province_names table displayed using DESCRIBE province_names.

3.3 Dataset Size

The record-count query returned 4,530 patients, 5,067 admissions, 27 doctors, and 13 province records. The column-count query confirmed 10 columns in patients, 5 in admissions, 4 in doctors, and 2 in province_names. Together, the four tables contain 21 columns.

	total_patients	total_admissions	total_doctors	total_provinces
▶	4530	5067	27	13

Figure 7. Total record count for all four tables.

	TABLE_NAME	total_columns
▶	admissions	5
	doctors	4
	patients	10
	province_names	2

Figure 8. Column count for each table.

3.4 Missing Value Identification

A NULL-value check was performed on the columns of each table. In the patients table, the output shows 2,059 NULL values in allergies and zero NULL values in the other displayed columns. The admissions, doctors, and province_names checks show zero NULL values in all columns included in their respective checks.

	null_patient_id	null_first_name	null_last_name	null_gender	null_birth_date	null_city	null_province_id	null_allergies	null_height	null_weight
▶	0	0	0	0	0	0	0	2059	0	0

Figure 9. NULL-value check for the patients table.

	null_patient_id	null_admission_date	null_discharge_date	null_diagnosis	null_attending_doctor_id
▶	0	0	0	0	0

Figure 10. NULL-value check for the admissions table.

Note: NULL Values in the Allergies Column
The **2,059 NULL values** indicate missing or unavailable allergy information; they do not confirm that a patient has no allergies. As required by **Question 5**, these values are to be replaced with **'NKA' (No Known Allergies)**. **NULL and NKA have different meanings.**

	null_doctor_id	null_first_name	null_last_name	null_specialty
▶	0	0	0	0

Figure 11. NULL-value check for the doctors table.

	null_province_id	null_province_name
▶	0	0

Figure 12. NULL-value check for the province_names table.

3.5 Duplicate Value Identification

Duplicate checks were performed on identifier fields in the patients, admissions, doctors, and province_names tables. The doctors and province_names duplicate-check outputs show no matching duplicate identifiers. The patient ID in the admissions table is a reference to the patients table and may be repeated when one patient has more than one admission; repeated patient IDs alone do not prove that the admission records themselves are duplicates.

	patient_id	total_count

Figure 13. Duplicate-value check output for the patient ID field.

	doctor_id	total_count

Figure 14. Duplicate-value check for doctor IDs.

	province_id	total_count

Figure 15. Duplicate-value check for province IDs.

Note: Repeated Patient IDs in the Admissions Table
In **patients**, **patient_id** uniquely identifies each patient. In **admissions**, it acts as a **Foreign Key**. Since one patient may have multiple admissions, repeated IDs are expected in this **One-to-Many Relationship** and do not necessarily indicate duplicate admission records.

4. SQL-Based Medical Data Analysis

The following analysis summarizes the results of the 34 SQL tasks. Each item explains what was retrieved or calculated and highlights the confirmed output.

Patient Information and Demographic Analysis (1–10)

1 | Male patient records

Retrieved **2,468 male patients** from the patients table.

2 | Patients with NULL allergy values

Identified **2,059 patients** whose allergy information is missing or unrecorded. NULL does not necessarily mean that a patient has no allergies.

3 | First names starting with C

Retrieved **302 patients** whose first names begin with the letter C.

4 | Patients weighing between 100 and 120

Retrieved **952 patients** whose recorded weight is between 100 and 120, including both limits.

5 | Replace NULL allergy values with NKA

The NULL allergy records were identified for the proposed replacement with NKA. The UPDATE operation was **not completed** because the database account did not have UPDATE permission; no successful data change is claimed.

6 | Combine first and last names

Combined first and last names into a full-name column for **4,530 patients** using CONCAT().

7 | Display full province names

Connected the patients and province_names tables to display each patient's name and full province name. The query returned **4,530 rows**.

8 | Patients born in 2010

Identified **55 patients** whose birth year is 2010.

9 | Patient with the greatest height

The tallest patient identified was **Sam Haruko**, with a recorded height of **226 cm**.

10 | Patients with selected patient IDs

Retrieved all available columns for the five selected patient IDs: **1, 45, 534, 879, and 1000**.

#	Time	Action		Message
1	20:33:13	SELECT	first_name, last_name, gender FROM patients WHERE gender = 'M' LIMIT 0, 50000	2468 row(s) returned
2	20:34:28	SELECT	first_name FROM patients WHERE first_name LIKE 'C%' LIMIT 0, 50000	302 row(s) returned
3	20:34:51	SELECT	first_name, last_name FROM patients WHERE weight BETWEEN 100 AND 120 LIMIT 0, 50...	952 row(s) returned
4	20:35:13	SELECT	CONCAT(first_name, ' ', last_name) AS full_name FROM patients LIMIT 0, 50000	4530 row(s) returned
5	20:35:49	SELECT	COUNT(*) AS total_patients FROM patients WHERE YEAR(birth_date) = 2010 LIMIT 0, 50000	1 row(s) returned
6	20:36:04	SELECT	first_name, last_name, height FROM patients WHERE height = (SELECT MAX(height) ...	1 row(s) returned
7	20:36:27	SELECT * FROM patients WHERE patient_id IN (1, 45, 534, 879, 1000) LIMIT 0, 50000		5 row(s) returned

MySQL Workbench action history showing returned row counts for selected queries.

Admission and Patient Condition Analysis (11–20)

11 | Total admission records

The admissions table contains **5,067 admission records**. This is the number of admission records, not the number of unique patients.

12 | Admissions with the same admission and discharge date

Retrieved **481 admission records** where the admission date and discharge date are the same.

13 | Admissions for patient ID 579

Identified **2 admission records** for patient ID 579.

14 | Unique cities in province NS

Retrieved **3 unique cities** in Nova Scotia (NS): **Halifax, Inverness, and Port Hawkesbury**.

total_admissions
5067

Output: total admission records = 5,067.

city
Halifax
Inverness
Port Hawkesbury

Output: patient ID 579 has 2 admissions.

	total_admissions
▶	2

Output: the three unique cities in province NS.

15 | Patients with height above 160 and weight above 70

Retrieved patients' first names, last names, and birth dates for records where height is greater than 160 and weight is greater than 70.

16 | Unique birth years in ascending order

Retrieved the different birth years recorded in the patients table and arranged them from the earliest year to the latest.

17 | First names that occur only once

Identified first names that appear exactly once in the patients table using GROUP BY and HAVING.

18 | Names starting and ending with s

Retrieved patient IDs and first names for names that start with s, end with s, and contain at least six characters.

19 | Patients diagnosed with Dementia

Connected patient and admission records to identify patients diagnosed with **Dementia**. The output contained **26 rows**.

20 | Sort first names by length

Sorted all patient first names by name length and then alphabetically. The query returned **4,530 rows**.

Gender, Repeat Admissions, and City Analysis (21–26)

21 | Male patient count

Identified **2,468 male patients** in the patients table.

	male_patients	female_patients
▶	2468	2062

Output: 2,468 male patients and 2,062 female patients.

22 | Female patient count

Identified **2,062 female patients**. The project statement repeats the combined gender-count task; the same output shows both totals.

23 | Patients admitted multiple times for the same diagnosis

Identified **11 patient-diagnosis combinations** with more than one admission.

24 | Patient count by city

Counted patients across **93 cities**. **Hamilton** had the highest count with **1,938 patients**, followed by **Toronto with 317**.

25 | Combine patients and doctors with their roles

Used UNION ALL to combine patient and doctor names and label each record as Patient or Doctor. The query returned **4,557 rows**.

26 | Allergies ordered by popularity

Counted non-NULL allergy values and arranged them from most common to least common. The output contained **63 allergy groups**.

Date, Formatting, Province, Weight, and Multi-Table Analysis (27–34)

27 | Patients born in the 1970s

Retrieved **623 patients** born between January 1, 1970 and December 31, 1979, arranged by birth date.

28 | Format names as LAST,FIRST

Displayed each name with the last name in **uppercase**, followed by a comma and the first name in **lowercase**. Results were sorted by first name in descending order.

29 | Province totals of patient height

Calculated total patient height by province. Provinces meeting the minimum total of 7,000 were **BC – 7,720**, **NS – 9,765**, and **ON – 678,037**.

	province_id	total_height
▶	BC	7720
	NS	9765
	ON	678037

Output: province-wise total patient height.

	weight_difference
▶	71

Output: weight difference for patients named Maroni = 71.

30 | Weight difference for patients named Maroni

The difference between the maximum and minimum recorded weights for patients with the last name Maroni was **71**.

31 | Admissions by day of the month

Counted admissions for every day of the month, from **1 to 31**, including days with no admissions, and arranged the results from the highest count to the lowest.

32 | Patients grouped by 10-unit weight ranges

Grouped patients into 10-unit weight ranges, counted the patients in each range, and sorted the groups in descending order.

33 | BMI-based obesity indicator

Calculated BMI using weight in kilograms and height in meters. The query displayed **1** when BMI was at least **30** and **0** otherwise. This threshold is the assumption used in the SQL analysis.

34 | Epilepsy patients attended by doctor Lisa

Identified **2 patients** diagnosed with Epilepsy who were attended by a doctor named Lisa: **Frank Anderson (Patient ID 468)** and **Precious Ashton (Patient ID 701)**. Both records showed the specialty **Obstetrician/Gynecologist**.

	patient_id	first_name	last_name	specialty
▶	468	Frank	Anderson	Obstetrician/Gynecologist
	701	Precious	Ashton	Obstetrician/Gynecologist

Output: two matching Epilepsy patient records and the attending doctor's specialty.

5. Power BI Data Model and Table Relationships

The Power BI data model shows how the four database tables—**patients**, **admissions**, **doctors**, and **province_names**—are connected. These relationships allow information from the different tables to be combined and analyzed in Power BI.

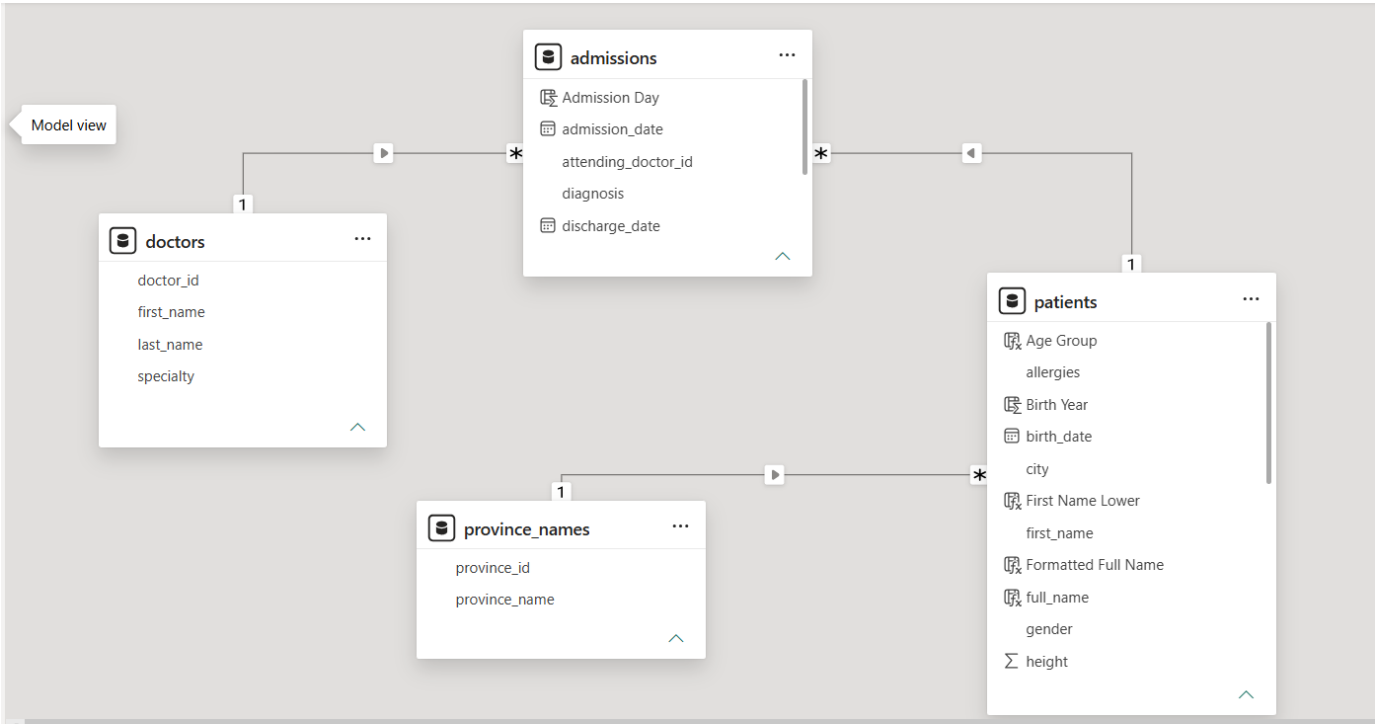


Figure: Power BI Data Model showing relationships between the four tables.

Relationships shown in the model

1. doctors → **admissions:** One doctor can be associated with multiple admission records.

2. patients → **admissions:** One patient can have multiple admission records.

3. province_names → **patients:** One province can be associated with multiple patients.

These are **One-to-Many Relationships**, with the '1' side representing the table containing the unique identifier and the '*' side representing the table that can contain multiple related records.